

Love of Variety, Habit Formation, and Dynamic Advertising

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Roadmap

- 1 Motivation
- 2 Model
- 3 Simulation Design
- 4 Results: Consumer Behavior
- 5 Next Steps

Motivation: Data Privacy, Ad Targeting, and Consumer Agency

- Concerns around data privacy and ad targeting
 - Facebook's $5B$ FTC settlement
 - Google ad tracking opt-out
- Amazon anti-trust scrutiny
 - Amazon accounts for almost 40% of US e-commerce sales (Statista, 2023)
 - Uses consumer data to target ads and build market share as both retailer and platform
- Consumer agency over data (opt out)
 - Apple iOS 14.5 update: 85% opt-out rate upon rollout (Flurry Analytics, 2022)
 - Facebook ad revenue down \$13bn in 2022 due to app tracking transparency (ATT) policy

Addressing the Concern

- How can we test the impact of policy interventions surrounding data tracking and ad targeting in a context like Amazon, where the firm has strong incentives to strategically target ads and promotions?
- Structural demand estimation is a powerful tool for policy analysis, but requires a model of consumer behavior and firm incentives

Key Challenge

How do we model consumer behavior in a way that captures the key drivers of consumption dynamics and firms' response?

Why Does Variety Matter?

- Canonical logit models ignore consumer preferences for variety, which are empirically important
- These preferences may shape consumer switching behavior, which in turn shapes firms' incentives to strategically target ads and promotions (i.e. to encourage switch to a higher markup product)
- Understanding the role of variety-seeking is valuable for modeling consumer behavior and evaluating the impact data privacy policies on both consumers and firms

Research Question

How much do consumer tastes for novelty and habit influence consumption dynamics and firms' advertising strategies in a dynamic market?

What This Paper Does

- ① **Model:** Derive a discrete-choice demand system where the utility of a product depends on both deviation from the consumer's choice history θ_{it} and product quality X_{jt}
- ② **Simulation:** Characterize consumer and firm behavior across preference regimes (β_i, γ_i) and advertising specifications
- ③ **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
- ④ **Counterfactuals (coming):** Study advertising targeting policy and welfare implications

Today's talk: Model + simulation results

Literature Connections

Demand side

- Lattin (1987): State dependence in consumer choice
- Chintagunta (1999): Extension to panel data
- Dubé, Hitsch, Rossi (2010): Habit formation and switching costs
- Bronnenberg, Dubé, Gentzkow (2012): Experience-based learning and consumer behavior

Advertising / firm side

- Shaffer and Zhang (1995): Theoretical basis for ad targeting in a two-period model
- Erdem & Keane (1996): Learning dynamics
- Zhang, Choi, Cheng (2025): Updated targeting game

Contribution

I build upon Chintagunta's dynamic demand model which micro-founds variety-seeking by using it to study how consumer preferences for novelty and habit shape consumption dynamics and firms' optimization strategies in a dynamic market.

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Consumer's Problem

Consumer i chooses product j in period t to maximize:

$$U_{ijt} = \underbrace{\beta_{ij} X_{jt}}_{\text{characteristic component}} + \underbrace{\gamma_i \log(1 + \Sigma_{ijt}^2)}_{\text{novelty component}} + \underbrace{\kappa a_{jt}}_{\text{advertising}} - \underbrace{\alpha p_{jt}}_{\text{price}} + \xi_j + \varepsilon_{ijt}$$

- where $\Sigma_{ijt} = X_{jt} - \theta_{it}$ is the deviation of product j 's characteristic from consumer i 's previously chosen characteristics θ_{it} in period t .
- at the moment, $\theta_{it} = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$, or the mean of previously chosen product characteristics.

Key parameters:

- $\beta_i > 0$: marginal utility of quality
- $\gamma_i > 0$: variety-seeking
- $\kappa > 0$: advertising shifter

Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$

State Variable: History of Choices

History of Choices θ_{it}

$$\theta_{it} = f(\{X_{js}\}_{s < t}) = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$$

The consumer's history, for simplicity, is summarized by the mean of previously chosen product characteristics. Updates each period as the consumer makes a new choice. This is the key state variable linking past behavior to current utility.

Initialization strategy:

- 1 **Initial state:** Simulate $T_{\text{prior}} = 5$ pre-sample periods from $J \in [1, 2, 3, 4, 5]$ to generate an experience-based prior.

Initial history yields more heterogeneous behavior — more on this shortly.

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Simulation Parameters

Table: Parameters

Param.	Value	Description
T	100	Periods
J	5	Products
S	1,000	Simulations
κ	0	Advertising (off)
α	0	Price (off)
ξ_j	0	Quality (off)
β_{df}	0	Discount factor (off)

Preference for Characteristic j :

- $\beta_i = 2$ — fixed quality sensitivity

Preference for Variety:

- $\gamma_i \in \{6, 9, 12\}$ — low, medium, high variety-seeking

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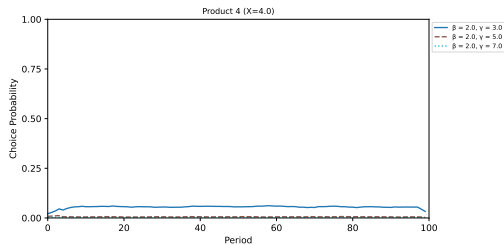
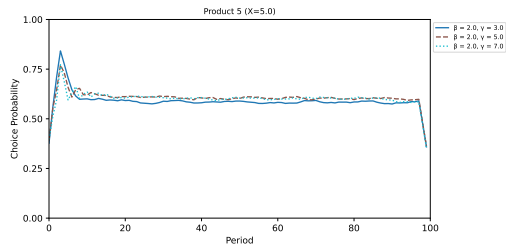
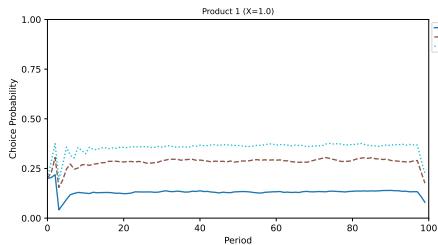
Naive Measurement

- Near static choice probabilities
- Any variation due to ε_{ijt} , not consumer preferences
- CCPs are not informative about consumer behavior or firm incentives
- Nearly all mass on Product 5, which has the highest quality X_{jt}

Table: Naive CCPs

Product	CCP
Product 1	0.02%
Product 2	0.54%
Product 3	4.05%
Product 4	22.3%
Product 5	72.0%

LOV Specification



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Road to Estimation

① Multiple consumers

Extend simulation to $I = 5$ consumers who have heterogeneous (β_i, γ_i) ; aggregate to market shares

② Consumer value function

Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values

③ Nielsen scanner data

Match simulated moments to data; identify (β_i, γ_i) using BLP instruments

④ Firm's Problem

Ad targeting policy; profit maximization; counterfactuals

Summary

- ① **Model:** Consumer utility depends on deviation from experiences; nests variety-seeking and habit formation
- ② **History:** Rich behavioral dynamics; regime differences generate cross-sectional sorting
- ③ **Next:** Multiple consumers, structural estimation, Nielsen data

Thank you — questions welcome

Appendix

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Simulation Algorithm (Consumer)

- ➊ Draw $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$ for all i, j, t
- ➋ Initialize $\bar{X}_{i0}$ (set mean or prior mean)
- ➌ For $t = 1, \dots, T$:
 - ➊ Compute $U_{ijt} = \beta_i X_{jt} - \gamma_i (X_{jt} - \bar{X}_{it})^2 + \kappa a_{jt} + \varepsilon_{ijt}$
 - ➋ Set $j^* = \arg \max_j U_{ijt}$
 - ➌ Update $\bar{X}_{it+1} = \frac{1}{t} \sum_{k \leq t} x_{ijk^*}$
- ➍ Compute CCPs via log-sum-exp for numerical stability:
 $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
- ➎ Average over $S = 1,000$ simulation draws